AI-Powered Thematic Analysis for Investment Research

Sponsor Organisation:	Sponsor Coordinator:	Project Mentor:
Business Domain: Investment Research Investment teams increasingly need to understand portfolio exposure to long-term structural themes such as AI infrastructure, energy transition, digital payments, or cybersecurity. Traditional sector classifications and stock-level thematic labels are often too broad and may not accurately reflect a company's exposure to a theme.	**Project Description:** Build an AI-powered thematic exposure estimation system. Core capabilities should include: *A. Theme definition engine* Allow users to define an investment theme with a template (e.g. theme description, included/excluded activities, positive/negative examples). *B. Theme taxonomy and sub-theme generation* Use AI to dynamically generate a hierarchy of sub-themes for each user-selected theme. *C. Company classification and exposure estimation* Extract relevant information (e.g. business segments, revenue streams) from publicly available sources; then classify the data into sub-theme categories and estimate exposure to the selected theme with confidence scores and supporting evidence. Where data is insufficient to support a classification, this should be identified. *D. Portfolio exposure mapping* Aggregate company-level outputs into portfolio and benchmark views. The system should identify the largest contributors to portfolio and active thematic exposures. (Optional stretch goal: Identify indirect beneficiaries in the value chain associated with the theme)	**Project Inputs:** • Publicly available information on companies, such as: annual reports and regulatory filings, investor presentations, company websites and business descriptions, earnings call transcripts, sustainability reports, news and company announcements, any revenue-segment disclosures and segment reporting.
Problem Statement: Existing thematic analyses rely on broad sector classification or binary thematic labels. This can overstate or understate the company's link to a theme, and does not consider indirect economic beneficiaries in the value chain. For example, growth in AI infrastructure may drive demand for Data Centres → Electricity Demand → Grid Expansion → Copper Demand → Mining Companies. The project explores whether AI can more accurately estimate a company's thematic exposure with greater granularity, and thereafter aggregate it into meaningful portfolio-level views.		**Project Deliverables:** • Theme definition and taxonomy engine capable of generating theme hierarchies and sub-theme structures from user-defined investment themes. • Company revenue exposure dataset showing theme/sub-theme exposures, confidence scores and supporting evidence (including links to sources for traceability). • Portfolio dashboard showing overall portfolio, benchmark and active exposures by theme and sub-theme, highlighting key contributors / detractors. • Evaluation and validation framework assessing classification accuracy and consistency, confidence calibration, portfolio-level reasonableness checks, taxonomy quality, and economic linkage plausibility. • Technical documentation and final presentation describing methodology, validation framework, limitations and future enhancements.